

TITLE PAGE

- Problem Statement ID-SIH26146

- Problem Statement Title-

AI-Powered Monitoring & Analysis of Bitcoin

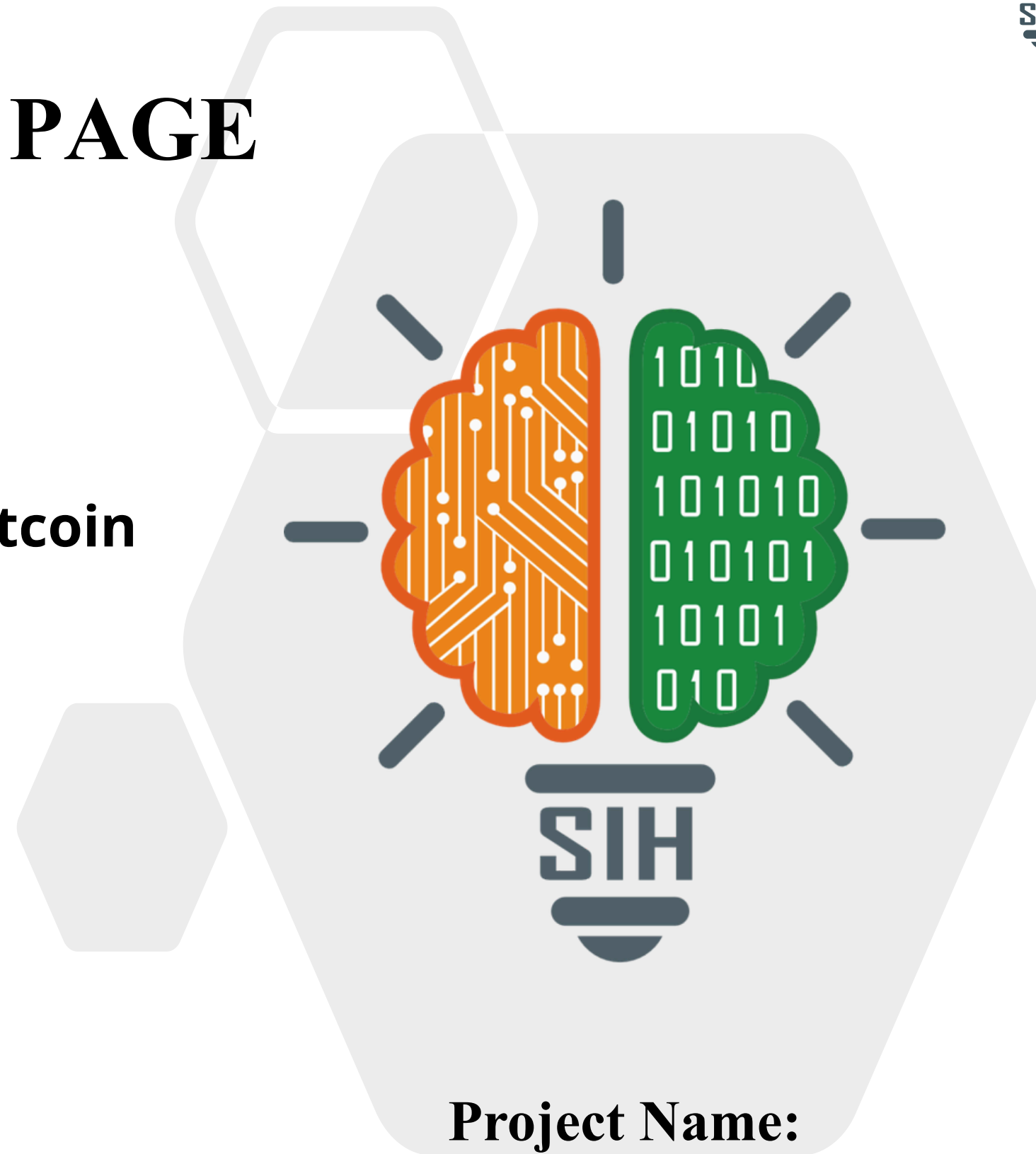
Transaction Traffic

- Theme-Blockchain & Cybersecurity

- PS Category-Software

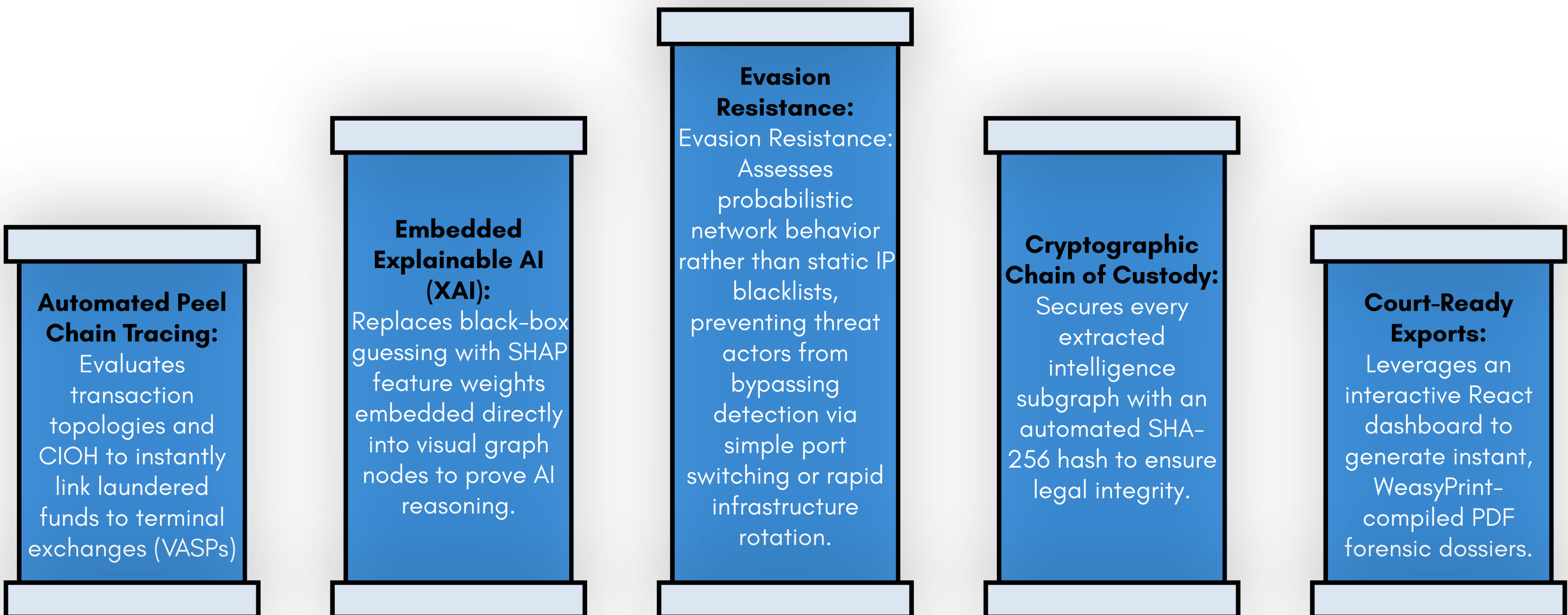
- Team ID-N/A

- Team Name-LocalDost



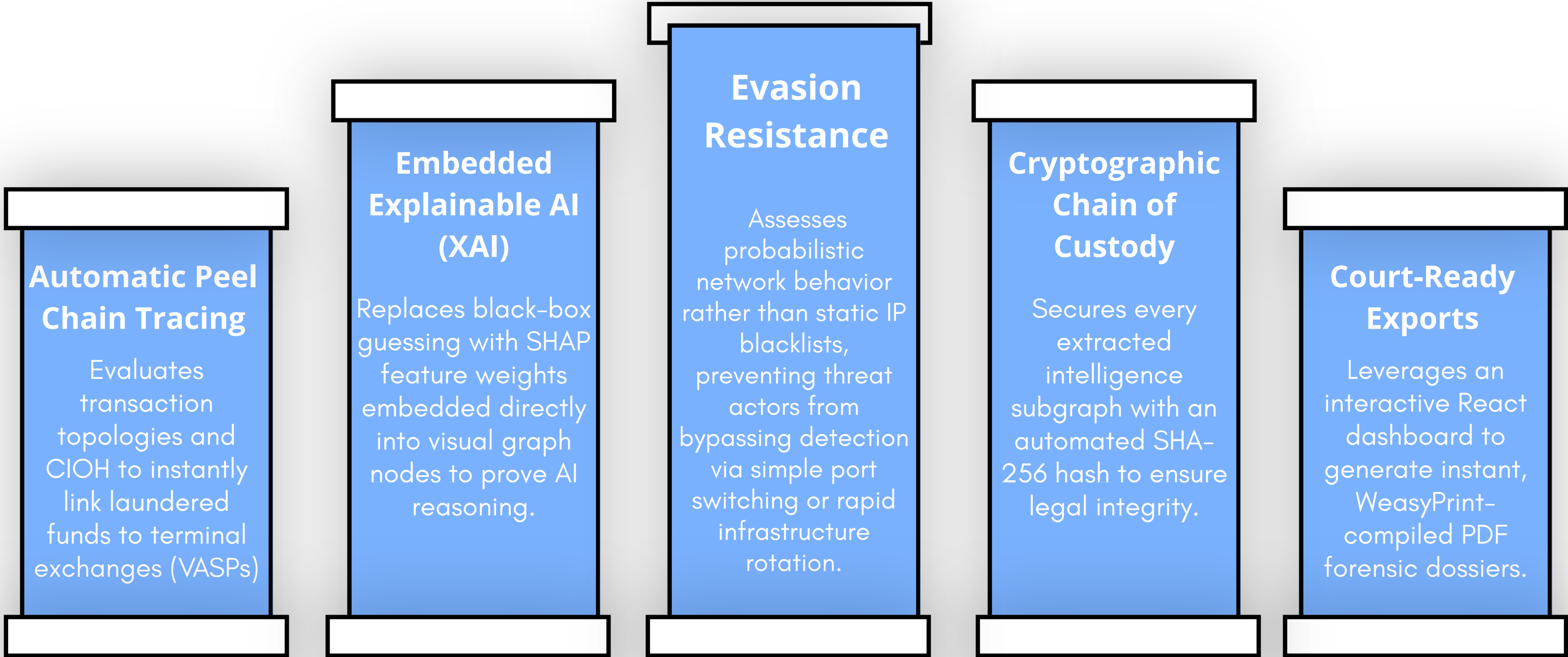
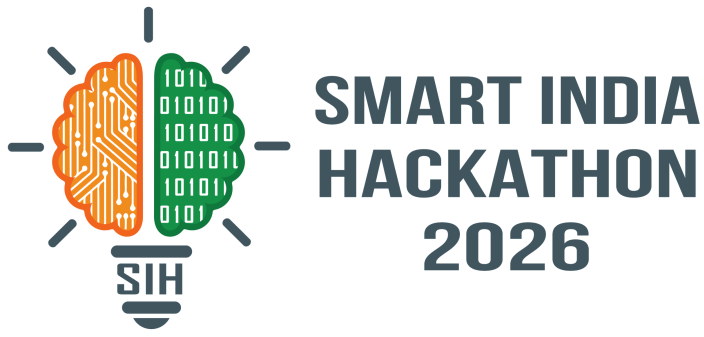
Project Name:
ShadowTrace-XAI

ShadowTrace-XAI: AI-Powered Bitcoin Transaction Forensics

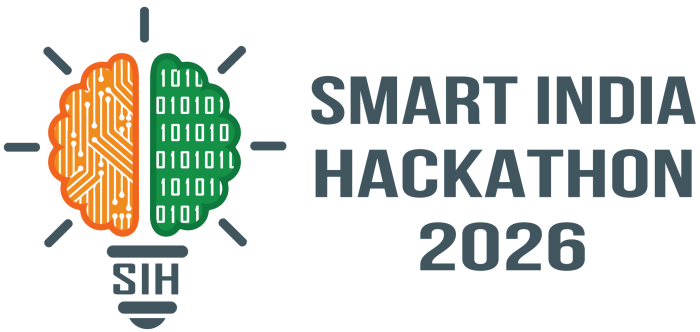




ShadowTrace-XAI: AI-Powered Bitcoin Transaction Forensics



TECHNICAL APPROACH



Process for Implementation

STAGE 1: Ingestion

Elliptic dataset, 166 features
Synthetic OSINT (TOR, ASN)
UTXO linking

STAGE 2: Backend AI

PyG graph from
nodes/edges.csv
GraphSAGE, weighted loss
GNExplainer

STAGE 3: Output

FastAPI
/api/graph/{txid}
Cytoscape dashboard
Hashed PDF dossier

LocalDost

AI/ML:
PyTorch, GraphSAGI

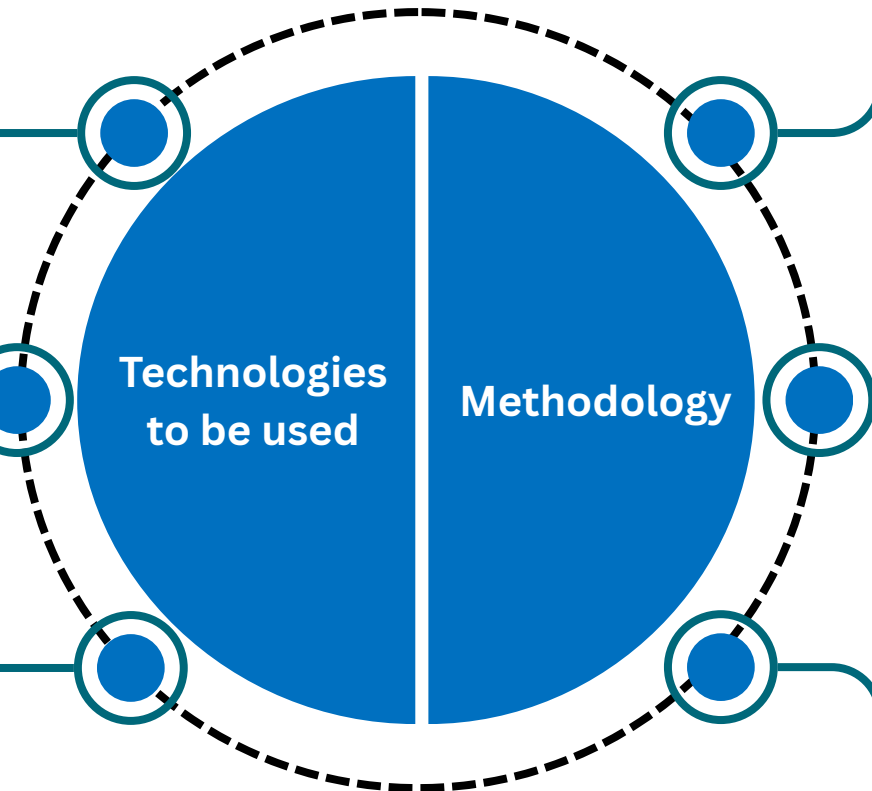
Data:
Pandas, GeoIP

Backend:
FastAPI, SHA-256

HARDWARE:
GPU Train, CPU deploy

LANGUAGES:
Python, TS

FRONTEND:
React, Cytoscape



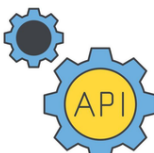
FEASIBILITY



Zero-Cost Infrastructure: Built on a fully open-source stack (PyTorch, FastAPI, React, WeasyPrint), eliminating enterprise licensing and heavy cloud costs.



Offline & Edge Deployment: Powered by a serverless SQLite database, allowing the active learning loop and forensic analysis to execute offline without networks.



Seamless System Integration: The lightweight REST API delivers JSON payloads, enabling rapid integration into law enforcement dashboards and legacy software.

PROBLEMS / RISKS

1) Graph scale & Latency
full-ledger embeddings,
memory bottleneck

2) Severe class imbalance
~2% illicit nodes,
misleading accuracy of the model

3) Feature drift & over-smoothing
Evasion tactics dilute detection signal

4) False-positive alert fatigue
High-recall burdens analysts,
stale static weights

OUR STRATEGIES

Neighbourhood subgraph sampling
Localised N-hop, minutes to milliseconds

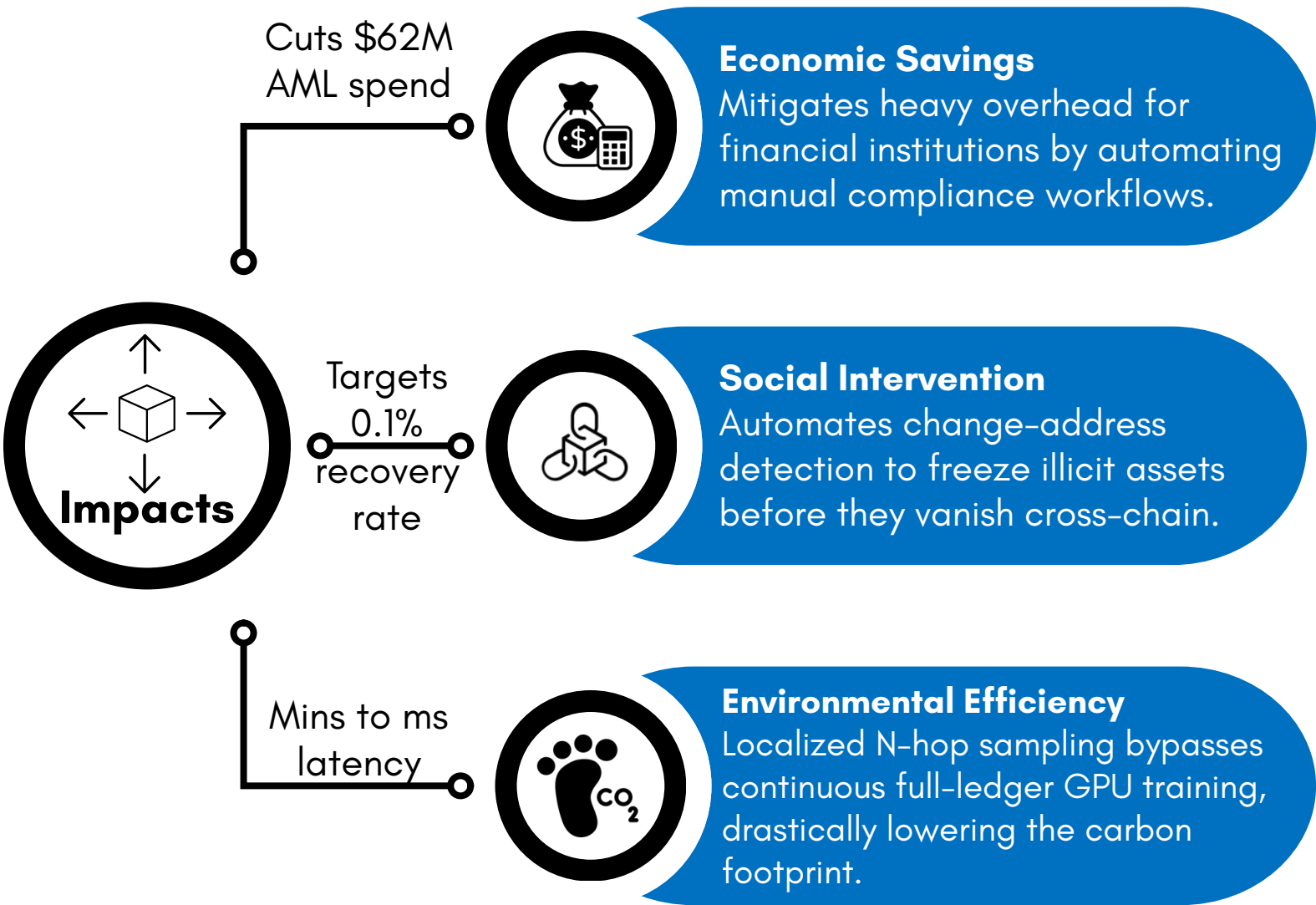
Weighted loss penalty
High pos_weight, optimizes F1 score

GraphNorm normalization
preserves variance across deep layers

Human-in-the-loop active learning
SQLite feedback drives continuous local retraining

By shifting the operational burden from manual analysis to automated Graph Neural Networks, ShadowTrace-XAI directly combats the **\$158 billion in global illicit crypto inflows** while aiming to dramatically improve the abysmal **0.1% global asset recovery rate**.

With **69% of cryptocurrency exchanges** currently failing **FATF Travel Rule compliance**, this tool transforms complex ledger data into actionable intelligence, ensuring seamless regulatory alignment and rapid disruption of malicious networks, protecting global markets from illicit financial flows.




Government Authorities

Empowers NTRO, FIU-IND, and CBI with rapid deanonymization and mathematically proven, court-ready WeasyPrint PDF evidence for prosecution.


VASPs & Institutions

Automates heavy compliance workflows, allowing financial entities to proactively flag sanctions evasion and illicit multi-hop flows in real time.


Citizens & Investors

Protects retail users by accelerating the disruption of ransomware, darknet, and fraud networks, fundamentally restoring trust in digital assets.

Academic & Technical Foundations

- Weber, M. et al. (2019): Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics
- Nakamoto, S. (2008): Bitcoin: A Peer-to-Peer Electronic Cash System
- Antonopoulos, A. M.: Mastering Bitcoin-UTXO lifecycle tracking and cryptographic parsing.
- PyTorch Geometric Documentation: Framework for our GraphSAGE architecture and XAI feature extraction.
- Meiklejohn, S. et al. (2013): A Fistful of Bitcoins: Characterizing Payments Among Men with No Names
- Ying, R. et al. (2019): GNNExplainer: Generating Explanations for Graph Neural Networks — Reference for the XAI framework used to extract interpretable subgraph structures and feature attributions for our dashboard.

Regulatory & Compliance Frameworks

- FATF (2021 Update): Updated Guidance for a Risk-Based Approach to Virtual Assets and Virtual Asset Service Providers
- FIU-IND (2026): PMLA AML/CFT Guidelines for Virtual Digital Assets